



Politecnico
di Torino

Algorithms and Programming

Iniziato martedì, 21 dicembre 2021, 17:13

Stato Completato

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Tempo impiegato 7 secondi

Valutazione 0,00 su un massimo di 9,00 (0%)

Domanda 1

Risposta non data

Punteggio max.:
2,00

The following recursive function is called by the main program with $n=3$, i.e., as $f(3)$.

```
int f (int n) {  
    if (n<=0) {  
        return (n);  
    }  
    return (f(n-1)+f(n-2)+f(n-3));  
}
```

Following the recursion tree, indicate the value returned by the function to the main program. Please, report your response as a unique integer value (with a sign, i.e., positive or negative). No other symbols must be included in the response. This is an example of the response format: -2

Risposta: ✖

La risposta corretta è : -7

Domanda 2

Risposta non data

Punteggio max.:
3,00

A matrix `mat` has `n` rows and `n` columns, and it stores float values.

Write a non-recursive function that receives the matrix `mat` and its size `n` as parameters, and it displays (on the standard output) all the values stored in the matrix (i.e., $[n \times n]$ float values) printing them with the order indicated by the following picture.

	0	1	2	3	...
0	1	2	4	7	11
1	3	5	8	12	...
2	6	9			
3	10				
...					

In other words, the numbers (1, 2, 3, etc.) reported in the picture represent the order in which the elements of the matrix must be displayed not their values. Thus, the function must display float values in the following order: `mat[0][0]`, `mat[0][1]`, `mat[1][0]`, `mat[0][2]`, `mat[1][1]`, etc.

The function prototype is the following:

```
int display (float **mat, int n);
```

Please, report the entire C code of the function. Briefly explain (in plain English language, within a C comment) the logic of the code.



Domanda 3

Risposta non data

Punteggio max.:
4,00

A list of lists represents the relationship between producers and products. Each producer's name is stored into an element of the main list and it has a related secondary list of products. Each product is characterized by an identifier (a string) and its price (a real value), and it is associated to an element of one secondary list.

Write the function to insert a new triple producer-product-price in this list of lists. Notice that the insertion procedure must discriminate among 3 cases:

1. The producer is not present in the database. In that case, the function must introduce the new producer with the new product and its price.
2. The producer is present but it does not have that product. In that case, the function must introduce the new product with its price in the secondary list associated with the right producer.
3. The producer and the product are already present. In this case, the function must update the price of that product for that producer.

The candidate can decide where to insert a new element (on top of the list, on the tail, or in a sorted position) for both the main and the secondary lists. The function must allocate all dynamically defined data fields.

The node data structure and the function prototypes are the following.

The function receives the reference to the pointer to the list of lists (head), and the triple producer name (name), product identifier (id), and price (price) to insert into the list of lists. Please, report the entire C code. Briefly explain (in plain English language, within a C comment) the logic of the code.

```
typedef struct producer_s producer_t;
typedef struct product_s product_t;
struct producer_s {
    char *name;
    producer_t *next;
    product_t *right;
};
struct product_s {
    char *id;
    float price;
    product_t *next;
};
```

```
void insert (producer_t **head, char *name, char *id, float price);
```

